Combustion Technology

(Lecture - 4)

⇒ Combustion

* combustion is a heterogeneous chemical reaction between substances, usually including oxygen and usually accompanied by generation of heat and light in the form of flame (Figure-1 A,B)
* the rate and speed at which the reactant combine is high
  - in part because of the nature of the chemical reaction itself, and
  - in part more energy is generated than can escape into the surroundings
  - the result is that the temperature of the reactant is raised to further accelerate and continue the reaction
* in simple word, it is a high temperature exothermic ~~reaction~~ rxn. between a fuel and atmospheric oxygen.

⇒ Bioresource Technology and Combustion

* the burning of biomass in the presence of air is used over a wide range of outputs to convert the chemical energy stored in biomass into heat, mechanical power and ~~ect~~ electricity
* process equipments such as stoves, furnaces, boilers, steam turbines turbogenerators, etc are used for the purpose
* hot gases (flue gases) at temperatures around 800-1000 °C are produced
* biomass with moisture content < 50% is used
* in case of moisture content > 50%, pre-drying is required
* combustion plants from very small scale (100 MW) to large scale (3000 MW) exist
* ~~now a~~ nowadays, due to higher conversion efficiency, co-combustion with coal in coal-fired power plants is an attractive option for utilization of biomass

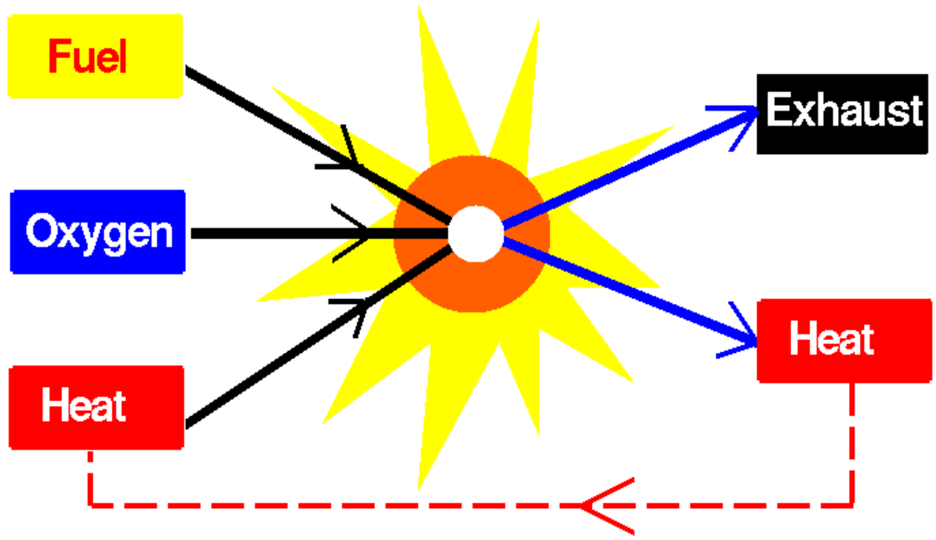


Figure-1A: Simple reperesentation of combustion

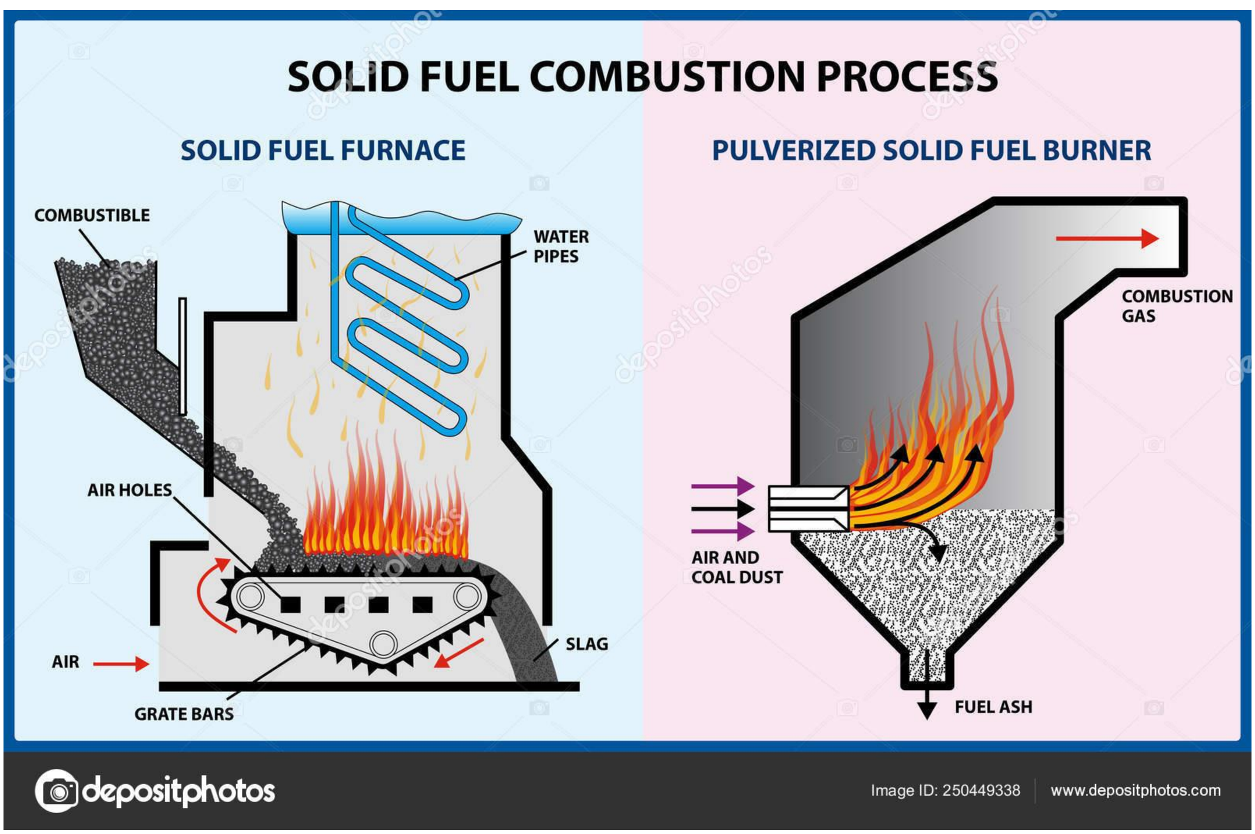


Figure-1B: Combustion of Biomass in a Furnace

* biomass combustion is related to significant pollutant formation due to two reasons:
  - incomplete combustion which leads to high ~~em~~ emissions of unburnt pollutants, such as CO, soot and polyaromatic hydrocarbons (PAH).
  - pollutants such as NO_x and particles are formed as a results of fuel constituents like N, K, Cl, Ca, Na, Mg, P and S.
* biomass furnaces exhibit relatively high emission of NO_x and submicron particles
* primary majors such as air and fuel stagings have been developed to reduce NOx reduction from 50% to 80%.
* efforts are always required for optimized plant operation in order to guarantee low emissions and high conversion efficiency.

⇒ Motivation for Energy from Biomass

* globally, there is increasing interest in biomass utilization for production of energy
* in most cases the driving force is – the CO_2 neutrality of sustainable cultivated biomass or the utilization of biomass residues and wastes
* combustion is the most important and mature ~~lest~~ technology available
* the availability of native and wastes biomass, ensures sustainable bioenergy utilization with the help of combustion technology
* improvements with respect to biomass conversion efficiency, reduction in pollutants emission and cost need to be further exploited.

⇒ Merits and Demerits of Combustion Technology.

* Merits
  - combustion technology based on renewable feed stock, like biomass would not run out

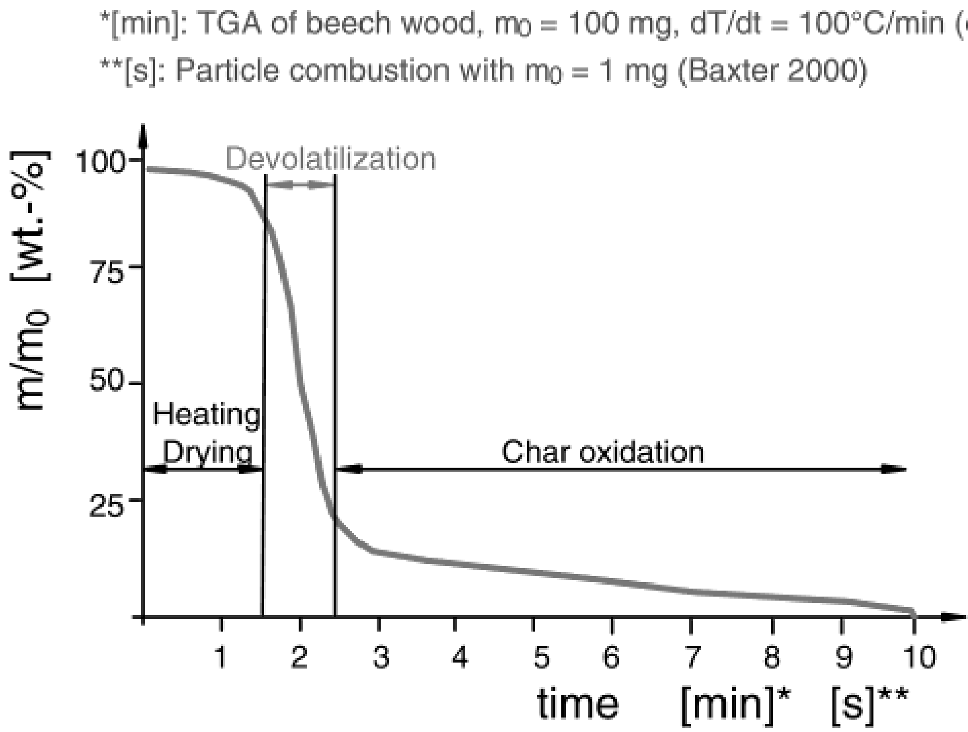
- lower reliance on foreign energy sources is another benifit of this technology based on biomass, as biomass is available every where
- combustion technology is the mature and commerically viable technology
- saves money as the feed stock (biomass) is cheaply available
- the heat produced in the process can directly be used for power generation

* Demerits
- due to emission of pollutants, it is not ecofriendly
- there may be unpredictable weather condition leading to draught and acute shortage of biomass feedstock for the power plant.
- No provision for storage of heat
- due to emission of large volume of gases, bigger equipment is required for cleaning

Note: When it comes to the renewable energy, the positives outweigh the negatives.

⇒ Fundamentals of Biomass Combustion

- a ~~complex~~ complex process which consists of consecutive heterogeneous and homogeneous chemical reactions (Figure-3).
- energy is liberated with the breakdown of C-C, C-H, C-O and O-H bonds in biomass
- chemical energy stored in biomass is converted into heat energy.
- the main process steps in combustion are: drying, devolatilization, gasification, char combustion and gas-phase oxidation.
- time used for each reaction depends on the fuel size and its properties, on temperature and on combustion conditions.



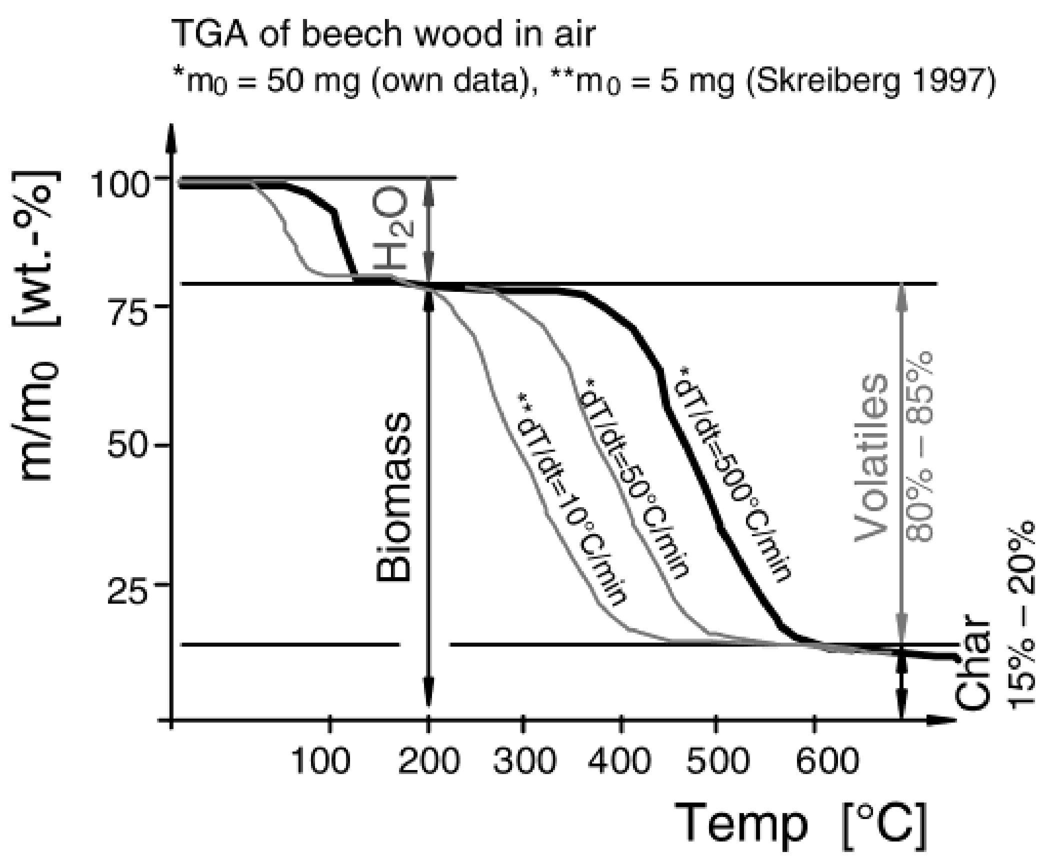


Figure-2: Mass loss as a function of time (above) and temperature (below) during combustion of wood in thermogravimetrical analysis (TGA). Source: Nussbaumer, 2003.

- batch combustion of small particles shows a distinct separation between volatile and char combustion phases with time (Figure 2)
- high contents of volatiles (80–85%) are respected for design of combustion appliances

* Excess Air Ratio, λ

- it is the main parameter in design of the system carrying combustion
- λ is defined as the ratio of available air (locally) to that of air required stoichiometrically

$$\lambda = \frac{\text{locally available air}}{\text{stoichiometric amount of air}}$$

- for typical biomass the combustion reaction may be described as

$$CH_{1.44}O_{0.66} + \lambda\, 1.03\,(O_2 + 3.76\,N_2)$$
$$\longrightarrow \text{intermediates } (C, CO, H_2, CO_2, C_mH_n, \text{etc.})$$
$$\longrightarrow CO_2 + 0.72\,H_2O + (\lambda - 1)\,O_2 + \lambda\, 3.87\,N_2 \quad - 439\ \text{kJ/mol}$$

where, $CH_{1.44}O_{0.66}$ represents the average composition of biomass, such as wood, straw, etc.

- in the above equation, biomass constituents, such as N, K, Cl, etc. have been neglected and the reactions have been shown in Figure – 3

* Pollutants in Biomass Combustion (Reference to Figure – 3)

1. Unburnt pollutants: CO, C_xH_y, PAH, tar, soot, unburnt carbon, H_2, HCN, NH_3, and N_2O
2. Pollutant from complete combustion: NO_x (NO and NO_2) CO_2, and H_2O, and
3. Ash and contaminants: ash particles (KCl, etc.), SO_2, HCl, PCDD/F, Cu, Pb, Zn, Cd, etc.

Note: the conversion of fuel nitrogen to NO_x has been reflected in Figure – 4

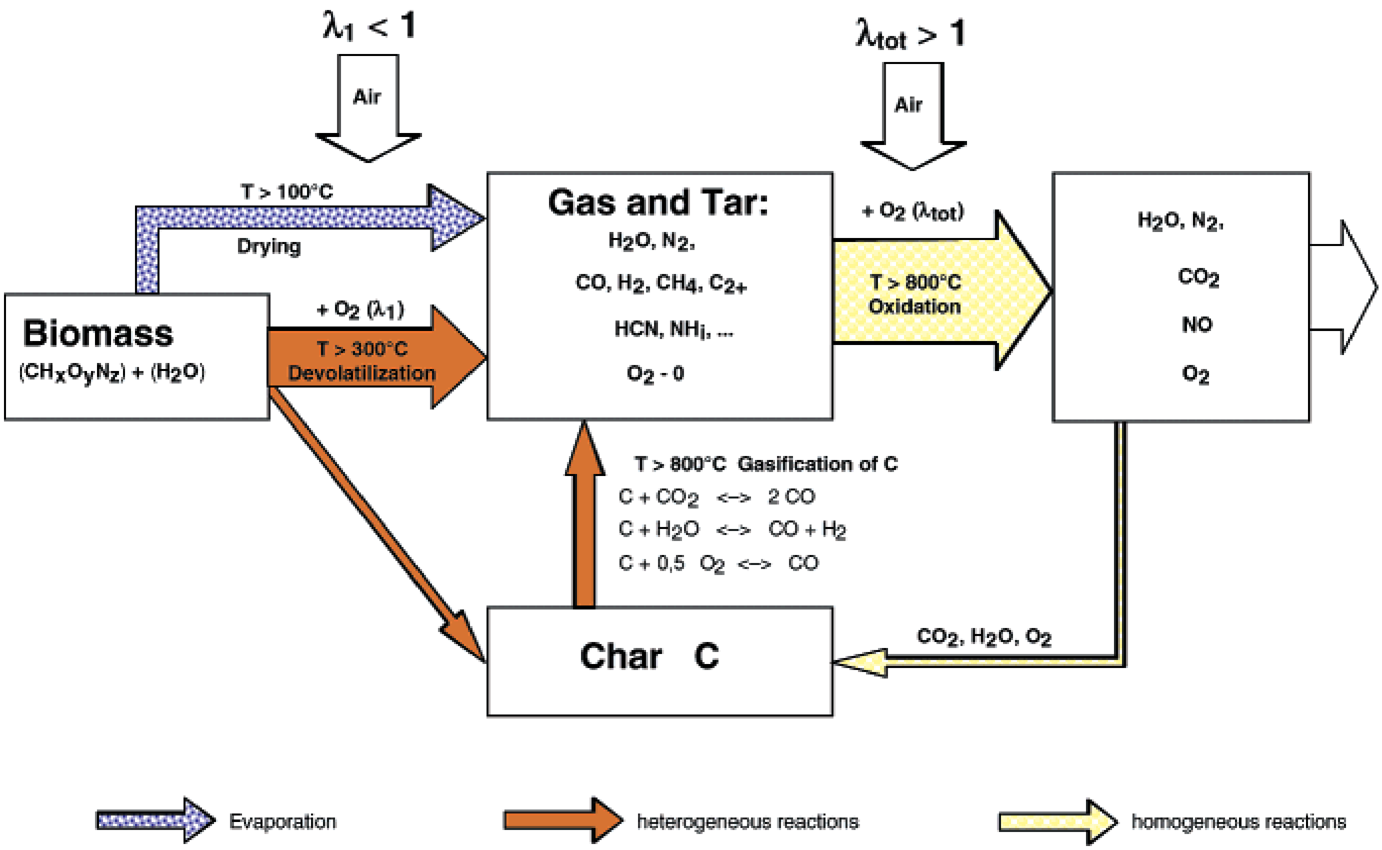
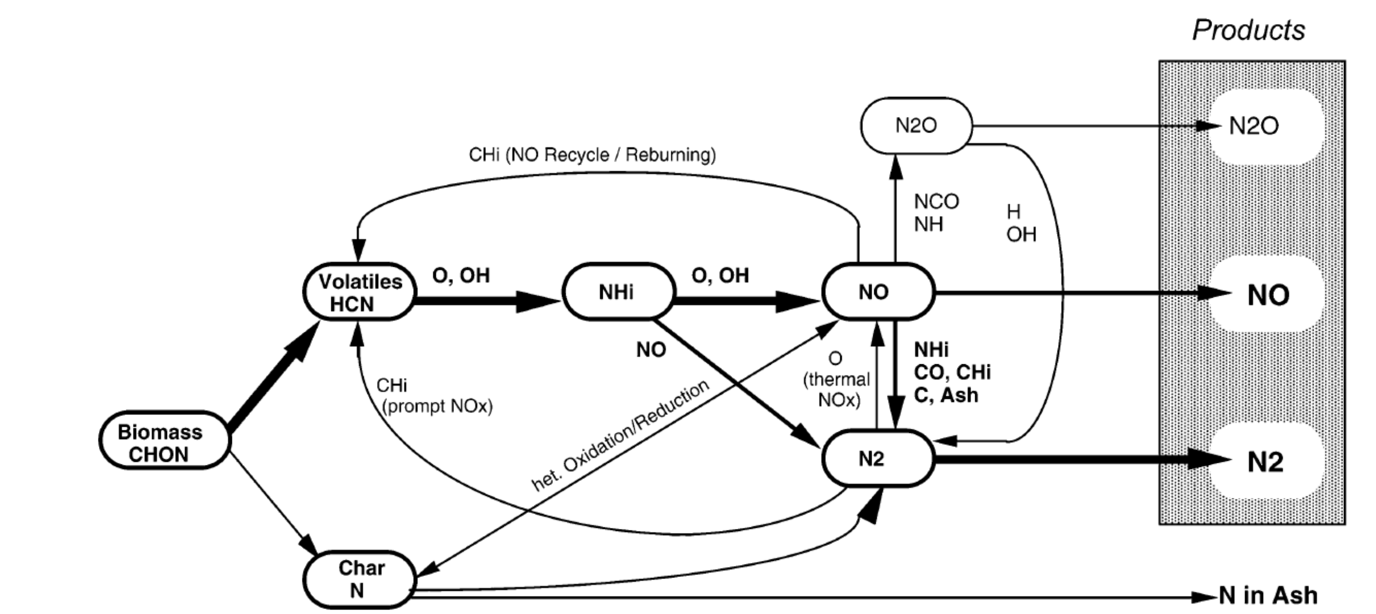


Figure-3: Main reactions during two-stage combustion of biomass with primary air and secondary air



Figutre-4 : Conversion of fuel nitrogen in biomass combustion.

* **Thermochemical Converters used in Biomass Combustion Technologies**

- the application of biomass is believed to play significant role in future in order to reduce GHG emissions
- three thermal conversion technologies, i.e. pyrolysis, gasification and combustion may be applied for the purpose
- combustion, being the most advanced and market-proven technology ~~may~~ play major role
- therefore, the application of biomass for energy production is primarily based on combustion
- combustion would be the dominant technology in future
- a broad spectrum of biomass combustion technologies using various types of biomass feedstocks, such as woody, harbaceous, agricultural and forestry wastes and residues are available
- the following three types of combustion technologies can be used for generation of energy from solid biomass (native, wastes and resides) as shown by figures 5A, 5B ~~may~~:
  - fixed bed combustion
  - fluidised bed combustion
  - pulverised fuel combustion

- fixed bed combustion includes
  - grate furnaces and underfeed stokers
  - primary air passes through a fixed bed where drying, gasification and oxidation of charcoal take place
  - the combustible gases are burnt by secondary air
  - combustion zone is separated from the fuel bed
  - it large flexibility with respect to moisture contents (10 – 60% by weight w.b.) and particle size (5 mm -> 10 cm)
  - mixing of wood fuels is possible but limited for harbaceous fuels.
  - nominal boiler capacity ranges from 10 kW to 50 kW.

⇒ fluidized bed combustion

- biomass fuel is burnt in suspension of gas and solid bed ~~mat~~ material where the primary air enters from the bottom
- based ~~on~~ on the fluidization velocity, bubbling fluidised bed and circulating fluidised bed may be differentiated
- in bubbling fluidised bed, the bed material is located in the bottom part of the furnace where the fluidisation velocity varies from 1.0 to 2.5 m/s.
- the circulating fluidised bed system is achieved by ~~increasin~~ increasing the velocity to 5 to 10 m/s by using smaller biomass fuel and sand particles
- the sand particles are carried with flue gas, separated in a hot cyclone or a U-beam separator, and fed back into the combustion chamber
- fluidized bed furnaces offers a large flexibility for moisture content from 10-50 wt.% w.b. and fuel mixing.
- low flexibility with respect to particle size
  - ✓ in BFB particle size < 80 mm
  - ✓ in CFB " " < 40 mm
- ~~low~~ low excess air ratios are possible with higher efficiency compensating the higher investment costs due to homogeneous combustion conditions
- boiler capacity may range from 20 MW to several 100 MW.

⇒ pulverized fuel combustion

- suitable for fuel with small particle size
- primary combustion air and fuel mixture are injected in the combustion chamber
- combustion takes place while the biomass particles are in suspension, and gas is burnt out after addition of secondary air.
- examples are: dust injection burners, muffle furnaces and cyclone burners.

W - moisture content usually below 20 wt.% w.b.
- particle size limited up to 20 mm
- a combination with grate and fluidised bed is possible
- most common for biomass co-firing, capacity of boiler varies from 500 kW to several 100 for co-firing systems;
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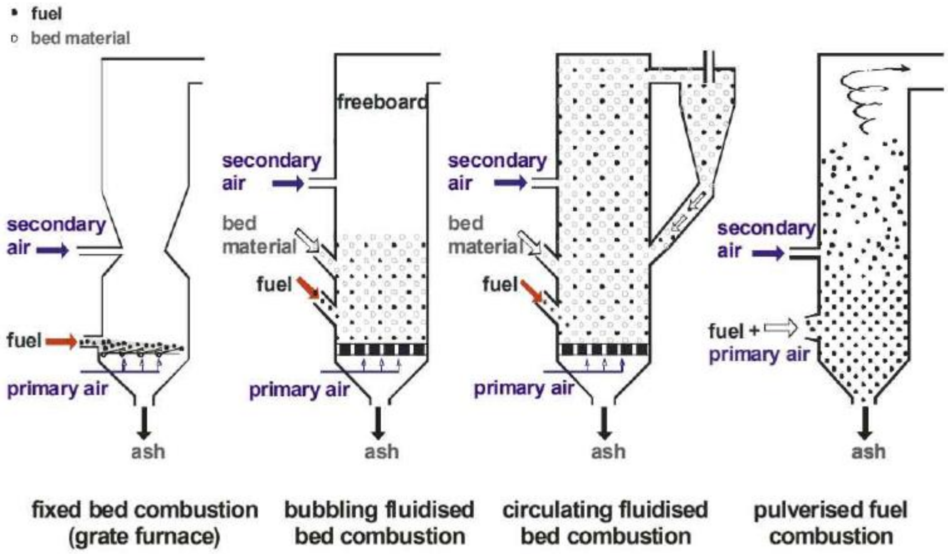


Figure-5: Different combustion technologies for biomass conversion to energy.

An overview of advantages, disadvantages, fields of application and capacity ranges of various combustion technologies have been reflected in Table-1.

Table-1: Overview of Advantages, Disadvantages and Fields of Applications of Various Combustion Technologies

Advantages	Disadvantages
Fixed bed combustion - capacity range: 100 kW – 50 MW	
large flexibility concerning moisture content (10 – 60 wt% w.b.) and particle size (5 - >100 mm) mixing of different wood fuels possible low investment costs for plants < 20 MW_{th} low operating costs less sensitive to slagging than fluidised bed furnaces	mixing of wood fuels and herbaceous fuels limited high excess oxygen (5 - 8 Vol%) decreases efficiency combustion conditions not as homogeneous as in fluidised bed furnaces
Fluidized bed combustion - capacity range: 20 MW – several 100 MW	
high flexibility concerning moisture content (10 – 55 wt% w.b.) and kind of biomass fuels used no moving parts in the hot combustion chamber high specific heat transfer capacity due to high turbulence low excess oxygen (1 - 4 vol%) raises efficiency and decreases flue gas flow	high investment costs, interesting only for plants > 20 MW_{th} high operating costs low flexibility with regard to particle size (< 80 mm for BFB, 40 mm for CFB) high dust load in the flue gas partial-load operation requires special technology high sensitivity concerning ash slagging
Pulverised fuel combustion - capacity range: 500 kW – several 100 MW (co-firing)	
Medium excess oxygen (4 - 6 Vol%) increases efficiency very good load control and fast alternation of load possible	water content usually < 20 wt% w.b. particle size of biofuel is limited (< 10-20 mm) high wear out of the insulation brickwork if cyclone or vortex burners are used an extra start-up burner is necessary

Samall Scale. Medium Scale and Large Size Biomass Combustion Plants

Concerning plant size, the different applications of biomass combustion can be divided into the following main fields:

- Small-scale biomass combustion systems: (capacity range: <100 kW_{th})
- Medium-scale combustion systems (capacity: range: 100 kW_{th} to 20 MW_{th})
- Large-scale combustion systems (capacity: range: >20 MW_{th})
- Co-firing of biomass in coal fired power stations (capacity range: some 100 MW_{th})

Small-scale biomass combustion units are mainly applied for residential heating systems. Here, different types of pellet burners, wood stoves and fire-place inserts as well as log wood, wood chips and pellets boilers are commonly used. These systems are only suitable for high quality wood fuels.

The **medium capacity range** covers biomass district and process heating and biomass combined heat and power (CHP) plants. Underfeed stokers, grate-fired furnaces and pulverised fuel burners are the main technologies applied in this capacity range. In Figures 6 examples for underfeed stokers and grate-fired furnaces are shown. Heat transfer is most commonly based on hot water boilers, but also steam boilers and thermal

oil boilers. These systems usually burn woody biomass fuels such as wood chips, sawdust, bark, forest residues and waste wood but also straw and other agricultural residues (e.g. sunflower husks). In addition to heat production for process and district heat supply, combined heat and power (CHP) systems are of increasing importance. A number of CHP technologies such as Stirling engines, the ORC process, steam engines and steam turbines can be combined with such plants.

The **large-scale range** (>20 MW_{th}) mainly comprises CHP plants and power plants with thermal capacities up to some 100 MW based on grate-fired and fluidised bed combustion systems (Figures -7 a, b) are usually fired with woody biomass fuels (wood chips, sawdust, bark, forest residues and waste wood) and straw but also with residues and wastes from the agricultural industry such as fruit stones, kernels, husks and shells. Usually steam boilers combined with steam turbines are applied.

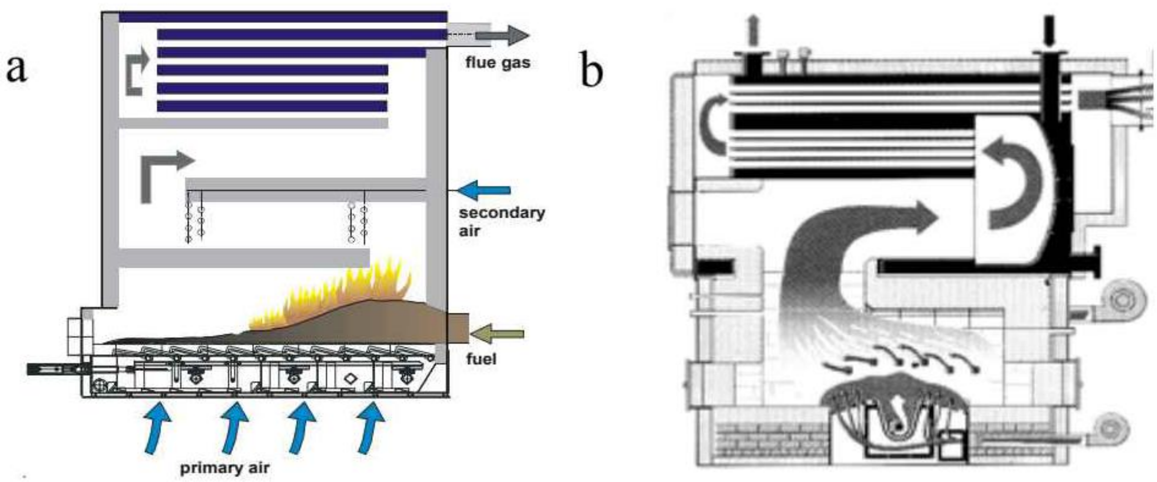
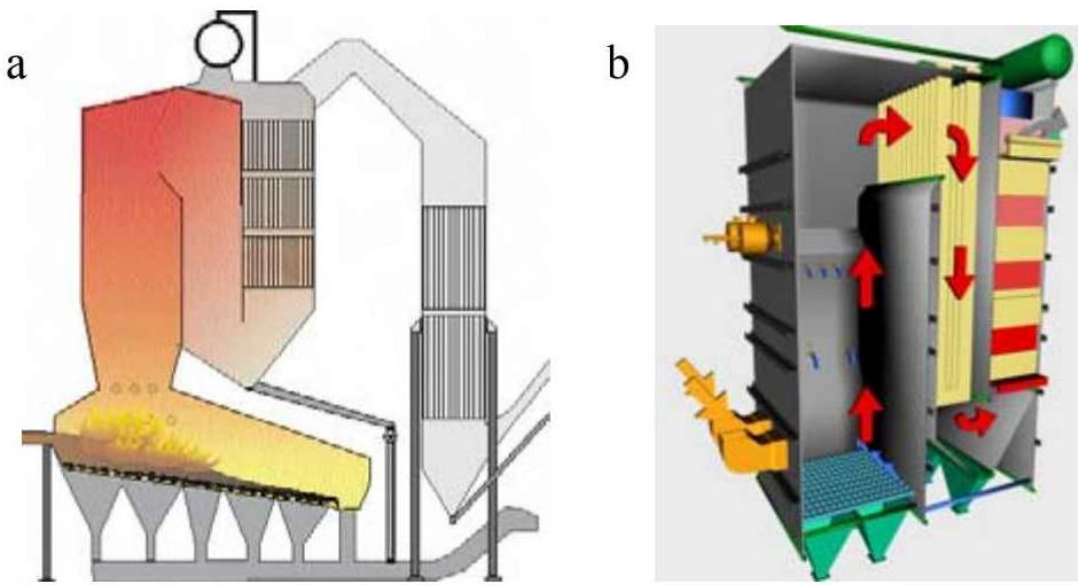


Figure-6: Medium scale biomass combustion plants (a: Grate-fired furnace, b:Underfeed stoker)



Figures-7: Large scale biomass combustion plants (a:grate-fired furnace, b:fluidized-bed)